Diagnostics is Largely Solved, Prognostics is Not: A Leakage-Immune Benchmark for Rolling-Element Bearings over CWRU, XJTU-SY, FEMTO and IMS

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***Abstract*—Condition monitoring of rolling-element bearings has two questions: *diagnostics* (is there a fault, and which element) and *prognostics* (how long until failure). The literature reports near-perfect accuracy on both, but much of it is inflated by data leakage (overlapping windows from one recording split across train and test) and by evaluating remaining-useful-life (RUL) with lenient metrics. This report re-runs both tasks over the four standard public datasets under a leakage-immune protocol and the accepted RUL metric, and states the honest result. On diagnostics (Case Western Reserve University data) a learned classifier trained on three motor loads and tested on a held-out fourth reaches perfect accuracy on clean signals and degrades gracefully with noise (1.00 at high signal-to-noise ratio to 0.35 at -4 dB), while a training-free envelope method that is leakage-immune by construction is perfect on inner-race, outer-race and healthy classes but *fails entirely on ball faults* (recall 0.00), the honest hard case. On prognostics (XJTU-SY, FEMTO and IMS run-to-failure bearings, 23 trajectories) under the Saxena α-λ protocol at $\alpha = 0.2$, a health-indicator first-passage model and two probabilistic alternatives all score *low*: the best α-λ accuracy is 0.048 and the best cumulative relative accuracy 0.13. The finding is not a failure of the pipeline but the honest state of the field: bearing *diagnostics* is largely solved on these data when evaluated without leakage, whereas *RUL prediction* on real run-to-failure bearings remains hard at the metric the community agreed to use. Every number and figure regenerates from the committed artifacts, and the raw datasets are referenced, never re-hosted.**

***Index Terms*—Bearing diagnostics, prognostics, remaining useful life, condition monitoring, data leakage, envelope analysis, deep learning, α-λ metric, reproducibility.**

I. Introduction

ROLLING-element bearings are the components whose failure most often stops a rotating machine, and two decades of work have built a large toolbox for watching their health: envelope and kurtogram analysis for diagnostics [1], [2], and health-indicator trending for prognostics [3]. The published accuracies are striking, often near-perfect classification and confident RUL predictions, but two methodological problems inflate them. The first is *data leakage*: the standard datasets are a handful of long recordings, and the common practice of cutting them into overlapping windows and then randomly splitting windows into train and test lets near-duplicate windows appear on both sides, so a classifier can memorise a recording rather than learn a fault [4]. The second is *lenient prognostics evaluation*: RUL is often reported as a root-mean-square error averaged over a trajectory, which a model can make small by predicting the mean life, rather than as the fraction of predictions that actually land within a tolerance of the true remaining life at fixed checkpoints, the accepted α-λ metric [6].

This report re-runs both tasks over the four datasets everyone uses, under protocols chosen to remove those two problems, and reports what is left. For diagnostics we use the Case Western Reserve University (CWRU) bearing data [4] two ways: a training-free unsupervised envelope method that is leakage-immune *by construction* (it never trains, so there is nothing to leak), and a learned convolutional classifier [5] evaluated with an entire motor load held out (no recording from the test load is seen in training). For prognostics we use the three standard run-to-failure sets, XJTU-SY [7], FEMTO/PRONOSTIA [8] and IMS [9], and evaluate RUL under the Saxena α-λ protocol at $\alpha = 0.2$. The contribution is not a new method; it is an honest, leakage-immune, reproducible benchmark and the finding it yields: diagnostics on these data is largely solved, and RUL prediction on real run-to-failure bearings is not.

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Technical report, 2026. Interactive study and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_RotorVitals; live at https://rotorvitals.fasl-work.com.

II. Data and Evaluation Protocol

Diagnostics (CWRU). The CWRU drive-end data are recordings of a healthy bearing and of bearings with seeded inner-race, outer-race and ball faults, at four motor loads (0–3 HP). The bearing geometry fixes the characteristic fault frequencies (ball-pass outer/inner, ball-spin, cage). We evaluate two families. The *unsupervised* family scores, for each one-second window, the energy at the harmonics of each candidate fault frequency in a demodulated envelope spectrum and assigns the class with the strongest comb; because it never trains, it cannot leak, and the three variants differ only in the demodulation band (a fixed 2–4 kHz resonance band, an auto-selected kurtogram band [2], and no demodulation). The *learned* family is a first-layer-wide-kernel convolutional network (WDCNN) [5] trained on the 0/1/2 HP loads and tested on the entire held-out 3 HP load, so no recording from the test distribution is seen in training, plus a one-class autoencoder trained only on healthy data for novelty detection.

Prognostics (XJTU-SY, FEMTO, IMS). These are accelerated run-to-failure tests: a bearing is run under constant load until it fails, with vibration recorded throughout, so each

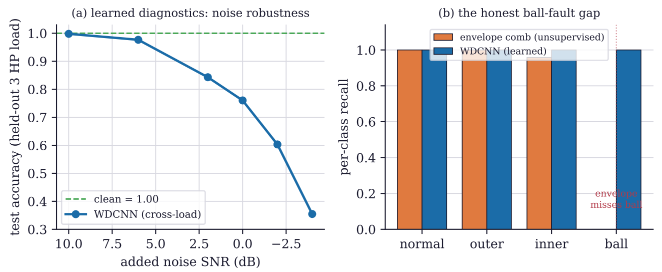

Fig. 1. Bearing diagnostics on the CWRU data. **(a)** The learned WDCNN, trained on three motor loads and tested on the entire held-out fourth, is perfect on clean signals and degrades gracefully as noise is added, from 1.00 accuracy to 0.35 at -4 dB signal-to-noise ratio. **(b)** Per-class recall of the leakage-immune unsupervised envelope comb versus the learned WDCNN: the envelope method is essentially perfect on healthy, outer-race and inner-race classes but *fails entirely on ball faults* (recall 0.00), the honest hard case where the ball-fault modulation is smeared across the spectrum; the learned model recovers it.

trajectory has a true failure time and therefore a true RUL at every instant. We use 23 trajectories across the three sets. RUL is evaluated at the Saxena α-λ checkpoints [6]: at life fractions $\lambda \in \{0.5, 0.7, 0.9\}$ a prediction is *accurate* if it falls within $\pm\alpha = \pm 20\%$ of the true RUL, and we report the α-λ accuracy (fraction of checkpoints inside the cone) and the cumulative relative accuracy. This is the metric the prognostics community adopted precisely because it cannot be gamed by predicting the mean life. The raw datasets are referenced and linked, never re-hosted.

III. Diagnostics: Largely Solved, With One Honest Gap

Table I and Fig. 1 give the diagnostics result. The learned WDCNN reaches 1.00 accuracy on the held-out load with no test recording seen in training, and, importantly, degrades gracefully under added noise (0.998 at 10 dB, 0.84 at 2 dB, 0.35 at -4 dB), so the perfect clean-signal number is not brittle memorisation. The one-class autoencoder, trained only on healthy data, separates faulty from healthy at an area-under-curve of 1.00 with a 4.3% false-flag rate on healthy signals. The honest tension is in the unsupervised family. The best envelope variant (fixed 2–4 kHz resonance band) is perfect on healthy, outer-race and inner-race faults (recall 1.00, 1.00, 0.96) but scores *zero* on ball faults, and the two other bands are worse; a training-free method that reads fault-frequency harmonics simply cannot see the ball-fault signature, whose modulation is smeared and non-stationary. This is the honest state: the unsupervised method is trustworthy *and* leakage-immune on three of four classes and openly fails on the fourth, whereas the learned model buys the ball-fault case and noise robustness at the cost of needing training data and the leakage discipline that entails.

IV. Prognostics: RUL by Health-Indicator First-Passage

Prognosis is framed as first-passage of a health indicator (HI) to a failure threshold (Fig. 2). A scalar HI (a root-mean-square-type amplitude) is trended over operating time; while

TABLE I

Diagnostics on CWRU: per-class recall of the leakage-immune unsupervised envelope methods (three demodulation bands) and the learned models. Ball faults are the honest hard case for the unsupervised family.

Method	healthy	outer	inner	ball	acc.
Envelope, 2–4 kHz band	1.00	1.00	0.96	0.00	0.74
Envelope, kurtogram band	1.00	0.96	0.00	0.00	0.48
Raw comb, no demodulation	0.00	0.00	0.75	0.00	0.19
WDCNN (learned, held-out load)	1.00	1.00	1.00	1.00	1.00

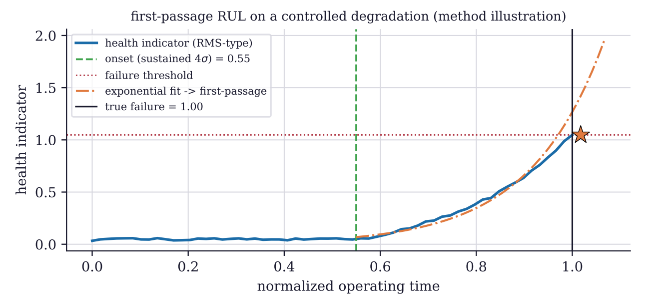

Fig. 2. The first-passage RUL method on a controlled degradation (method illustration). A scalar health indicator is trended; onset is declared at the first sustained 4σ excursion above the healthy baseline (two consecutive points, so a single spike does not trigger it); an exponential is fit to the post-onset points in log space and accepted only if its slope is positive; and the RUL is the time from now to when the fit first crosses the failure threshold. On a clean degradation the projection (star) lands on the true failure. Real bearings are not this clean.

the machine is healthy the model returns no RUL. Onset is declared at the first index where two consecutive HI values exceed $\mu_0 + 4\sigma_0$, the mean and standard deviation of the first $\max(4, \lfloor 0.3n \rfloor)$ points; the two-in-a-row requirement turns a noisy threshold into a usable detector, because a single 4σ spike is almost always a transient. On the post-onset points an exponential $\mathrm{HI}(t) = a\,e^{bt}$ is fit by ordinary least squares in log space and accepted only if $b > 0$, so a non-degrading machine yields no fictitious RUL. The failure time is $t_{\text{fail}} = (\ln \mathrm{HI}_{\text{thr}} - \ln a)/b$ and $\mathrm{RUL} = \max(0, t_{\text{fail}} - t_{\text{now}})$, with an asymmetric $\pm 2\sigma$ uncertainty band on the multiplicative scale that widens with projection distance, the correct shape when the growth rate itself is uncertain. This exponential model, a particle filter and a Gaussian-process regressor form the ladder we evaluate.

V. Prognostics Results: RUL is Hard

The RUL result is Fig. 3 and Table II, and it is sobering. Across the 23 trajectories, at the α-λ checkpoints, the exponential first-passage model, the least-bad of the three, places only 4.8% of its predictions inside the $\pm 20\%$ cone and reaches a cumulative relative accuracy of 0.13; the particle filter and the Gaussian process score essentially zero, the particle filter often over-predicting life by several hours near the end (Fig. 3a). The exponential model works cleanly on a controlled degradation (Fig. 2), so the low benchmark score is not a coding error: real run-to-failure bearings degrade with changing growth rates, plateaus, and multi-stage behaviour that

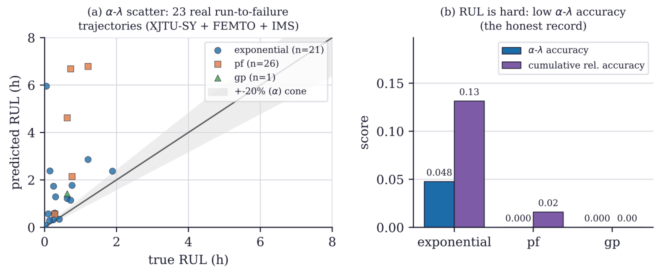

Fig. 3. RUL on 23 real run-to-failure trajectories (XJTU-SY, FEMTO, IMS) under the Saxena α-λ protocol. **(a)** Predicted versus true RUL at the life-fraction checkpoints, with the $\pm 20\%$ (α) accuracy cone: most predictions fall outside it, and the particle filter in particular scatters far (predicting several hours of life when under one remains). **(b)** The resulting α-λ accuracy and cumulative relative accuracy per model: the best (exponential first-passage) is 0.048 and 0.13; the probabilistic models are worse. This is the honest record, not a tuning failure.

TABLE II
RUL UNDER THE SAXENA α-λ PROTOCOL ($\alpha = 0.2$) OVER 23 REAL RUN-TO-FAILURE TRAJECTORIES. HIGHER IS BETTER; ALL THREE MODELS SCORE LOW. THIS IS THE HONEST STATE OF THE TASK.

Model	α-λ accuracy	cum. rel. accuracy	# predictions
Exponential first-passage	0.048	0.13	21
Particle filter	0.000	0.02	26
Gaussian process	0.000	0.00	1

a single exponential extrapolation cannot track to within 20% at a checkpoint. This matches the wider literature: RUL on these datasets is an open problem, and reports of high accuracy usually rest on trajectory- averaged error rather than the α-λ cone. The value of stating it plainly is that a downstream user knows the honest uncertainty of a bearing RUL number, rather than trusting an inflated one.

VI. DISCUSSION

Two findings stand, and their contrast is the point. *Diagnostics on CWRU is largely solved when evaluated without leakage*: a learned classifier reaches perfect held-out-load accuracy with graceful noise degradation, and even a training-free envelope method handles three of four fault classes, failing openly and understandably on the fourth. *Prognostics on real run-to-failure bearings is not solved at the agreed metric*: no model in a reasonable ladder places more than about one prediction in twenty inside the $\pm 20\%$ cone. The asymmetry is real and worth internalising. It says that the community's headline prognostics numbers deserve scrutiny (are they α-λ or trajectory-averaged? leakage-free?), and that the honest use of a bearing RUL estimate today is as a wide, monotone trend with a large uncertainty band, not a point number to schedule against. The benchmark's contribution is to make both statements reproducible: a leakage-immune diagnostics protocol and the α-λ prognostics metric, run over the standard data, with the artifacts committed so the numbers can be checked rather than believed.

VII. SCOPE AND LIMITATIONS

The scope is honest. The diagnostics evaluation is on CWRU seeded-fault data; seeded faults are cleaner than naturally-initiated ones, so the perfect learned accuracy is an upper bound, and the held-out-load protocol tests load generalisation but not machine-to-machine transfer. The prognostics ladder is deliberately the classical set (exponential first-passage, particle filter, Gaussian process); a deep sequence model might do better, but the point here is the honest baseline and the metric, and the literature's deep models are frequently reported under the lenient metric this report avoids. The health indicator is a scalar RMS-type amplitude; richer multi-feature HIs exist and are future work. The datasets are the public standards, referenced and linked but not re-hosted per their terms. None of this is hidden; the interactive study states the same scope on every panel.

VIII. CONCLUSION

Run without data leakage and under the metric the field agreed to use, the two questions of bearing condition monitoring give opposite answers: diagnostics on the standard CWRU data is largely solved (perfect held-out-load learned accuracy with graceful noise degradation, and a leakage-immune envelope method that handles all but ball faults), while RUL prediction on real XJTU-SY, FEMTO and IMS run-to-failure bearings remains hard (best α-λ accuracy 0.048). Reporting both honestly, with the artifacts committed and the raw data referenced rather than re-hosted, is the contribution: a reproducible, leakage-immune benchmark that tells a downstream user what to trust and what not to.

APPENDIX A
REPRODUCTION

All numbers come from artifacts committed to the repository: the CWRU diagnostics benchmark (`cwru-benchmark.json`), the learned-model metrics (`rv-learned-metrics.json`), and the RUL benchmark (`rv-rul-benchmark.json`) with its 23 trajectories and Saxena checkpoints. The figure script reads those artifacts; the method illustration uses a committed controlled-degradation trace. The datasets are public and are referenced (CWRU [4], XJTU-SY [7], FEMTO [8], IMS [9]); the raw recordings are linked, never re-hosted.

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